



National
Qualifications
2016

X747/76/11

**Mathematics
Paper 1
(Non-Calculator)**

THURSDAY, 12 MAY

9:00 AM – 10:10 AM

Total marks — 60

Attempt ALL questions.

You may NOT use a calculator.

Full credit will be given only to solutions which contain appropriate working.

State the units for your answer where appropriate.

Answers obtained by readings from scale drawings will not receive any credit.

Write your answers clearly in the spaces provided in the answer booklet. The size of the space provided for an answer should not be taken as an indication of how much to write. It is not necessary to use all the space.

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FORMULAE LIST

Circle:

The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle centre $(-g, -f)$ and radius $\sqrt{g^2 + f^2 - c}$.

The equation $(x - a)^2 + (y - b)^2 = r^2$ represents a circle centre (a, b) and radius r .

Scalar Product:

$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$

or $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$ where $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$.

Trigonometric formulae:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$= 1 - 2 \sin^2 A$$

Table of standard derivatives:

$f(x)$	$f'(x)$
$\sin ax$	$a \cos ax$
$\cos ax$	$-a \sin ax$

Table of standard integrals:

$f(x)$	$\int f(x)dx$
$\sin ax$	$-\frac{1}{a} \cos ax + c$
$\cos ax$	$\frac{1}{a} \sin ax + c$

Attempt ALL questions

Total marks – 60

1. Find the equation of the line passing through the point $(-2, 3)$ which is parallel to the line with equation $y + 4x = 7$.

2

2. Given that $y = 12x^3 + 8\sqrt{x}$, where $x > 0$, find $\frac{dy}{dx}$.

3

3. A sequence is defined by the recurrence relation $u_{n+1} = \frac{1}{3}u_n + 10$ with $u_3 = 6$.

(a) Find the value of u_4 .

1

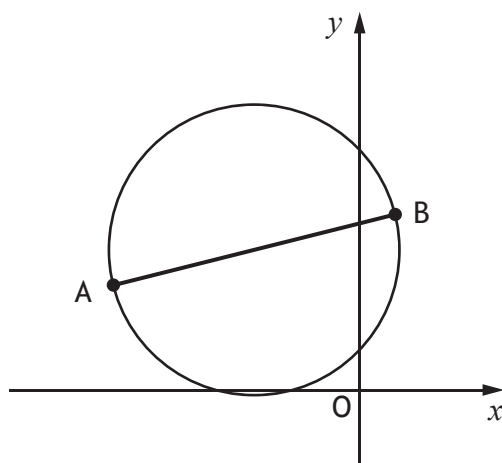
(b) Explain why this sequence approaches a limit as $n \rightarrow \infty$.

1

(c) Calculate this limit.

2

4. A and B are the points $(-7, 3)$ and $(1, 5)$.
AB is a diameter of a circle.



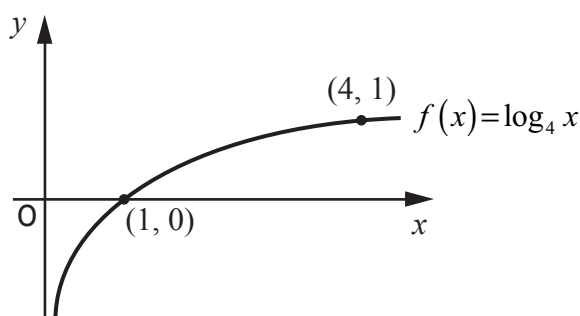
Find the equation of this circle.

3

5. Find $\int 8\cos(4x+1)dx$. 2
6. Functions f and g are defined on $\mathbb{R}$, the set of real numbers.
The inverse functions f^{-1} and g^{-1} both exist.
- (a) Given $f(x) = 3x + 5$, find $f^{-1}(x)$. 3
- (b) If $g(2) = 7$, write down the value of $g^{-1}(7)$. 1
7. Three vectors can be expressed as follows:
- $$\begin{aligned}\vec{FG} &= -2\mathbf{i} - 6\mathbf{j} + 3\mathbf{k} \\ \vec{GH} &= 3\mathbf{i} + 9\mathbf{j} - 7\mathbf{k} \\ \vec{EH} &= 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}\end{aligned}$$
- (a) Find $\vec{FH}$. 2
- (b) Hence, or otherwise, find $\vec{FE}$. 2
8. Show that the line with equation $y = 3x - 5$ is a tangent to the circle with equation $x^2 + y^2 + 2x - 4y - 5 = 0$ and find the coordinates of the point of contact. 5

9. (a) Find the x -coordinates of the stationary points on the graph with equation $y = f(x)$, where $f(x) = x^3 + 3x^2 - 24x$. 4
- (b) Hence determine the range of values of x for which the function f is strictly increasing. 2

10. The diagram below shows the graph of the function $f(x) = \log_4 x$, where $x > 0$.



The inverse function, f^{-1} , exists.

On the diagram in your answer booklet, sketch the graph of the inverse function. 2

11. (a) A and C are the points $(1, 3, -2)$ and $(4, -3, 4)$ respectively.
Point B divides AC in the ratio 1 : 2.
Find the coordinates of B. 2

- (b) $k\vec{AC}$ is a vector of magnitude 1, where $k > 0$.

Determine the value of k . 3

12. The functions f and g are defined on $\mathbb{R}$, the set of real numbers by

$$f(x) = 2x^2 - 4x + 5 \text{ and } g(x) = 3 - x.$$

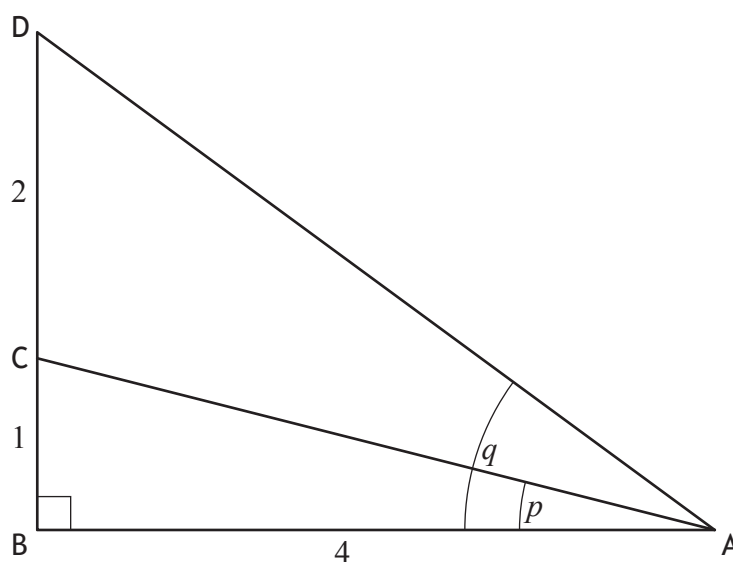
(a) Given $h(x) = f(g(x))$, show that $h(x) = 2x^2 - 8x + 11$.

2

(b) Express $h(x)$ in the form $p(x+q)^2 + r$.

3

13. Triangle ABD is right-angled at B with angles $BAC = p$ and $BAD = q$ and lengths as shown in the diagram below.



Show that the exact value of $\cos(q-p)$ is $\frac{19\sqrt{17}}{85}$.

5

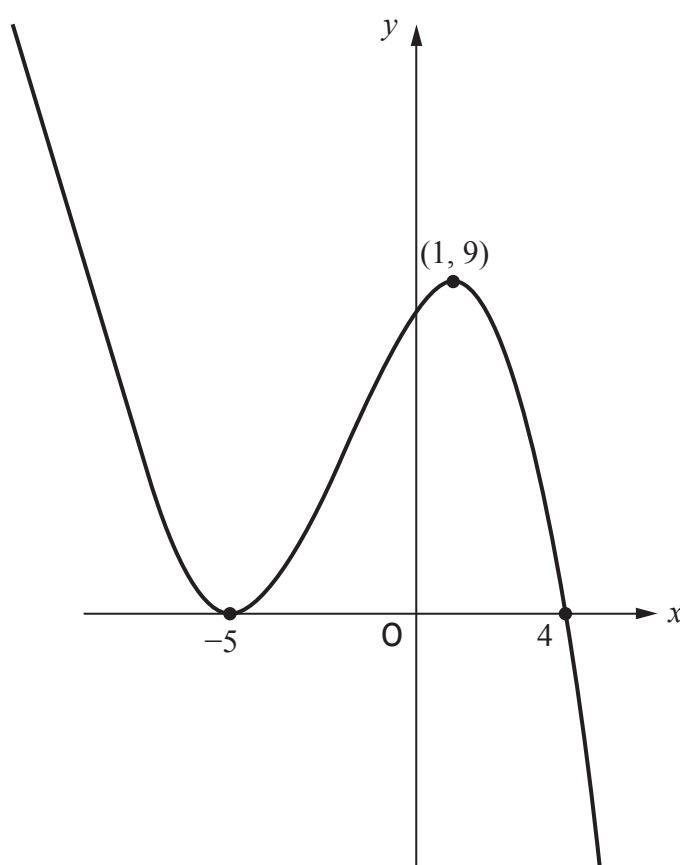
14. (a) Evaluate $\log_5 25$.

1

(b) Hence solve $\log_4 x + \log_4 (x - 6) = \log_5 25$, where $x > 6$.

5

15. The diagram below shows the graph with equation $y = f(x)$, where $f(x) = k(x - a)(x - b)^2$.



(a) Find the values of a , b and k .

3

(b) For the function $g(x) = f(x) - d$, where d is positive, determine the range of values of d for which $g(x)$ has exactly one real root.

1

[END OF QUESTION PAPER]

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National
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X747/76/01**Mathematics Paper 1 (Non-Calculator)
Answer Booklet**

THURSDAY, 12 MAY

9:00 AM – 10:10 AM



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Full name of centre

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Surname

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Month

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Year

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Scottish candidate number

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Write your answers clearly in the spaces provided in this booklet. The size of the space provided for an answer should not be taken as an indication of how much to write. It is not necessary to use all the space.

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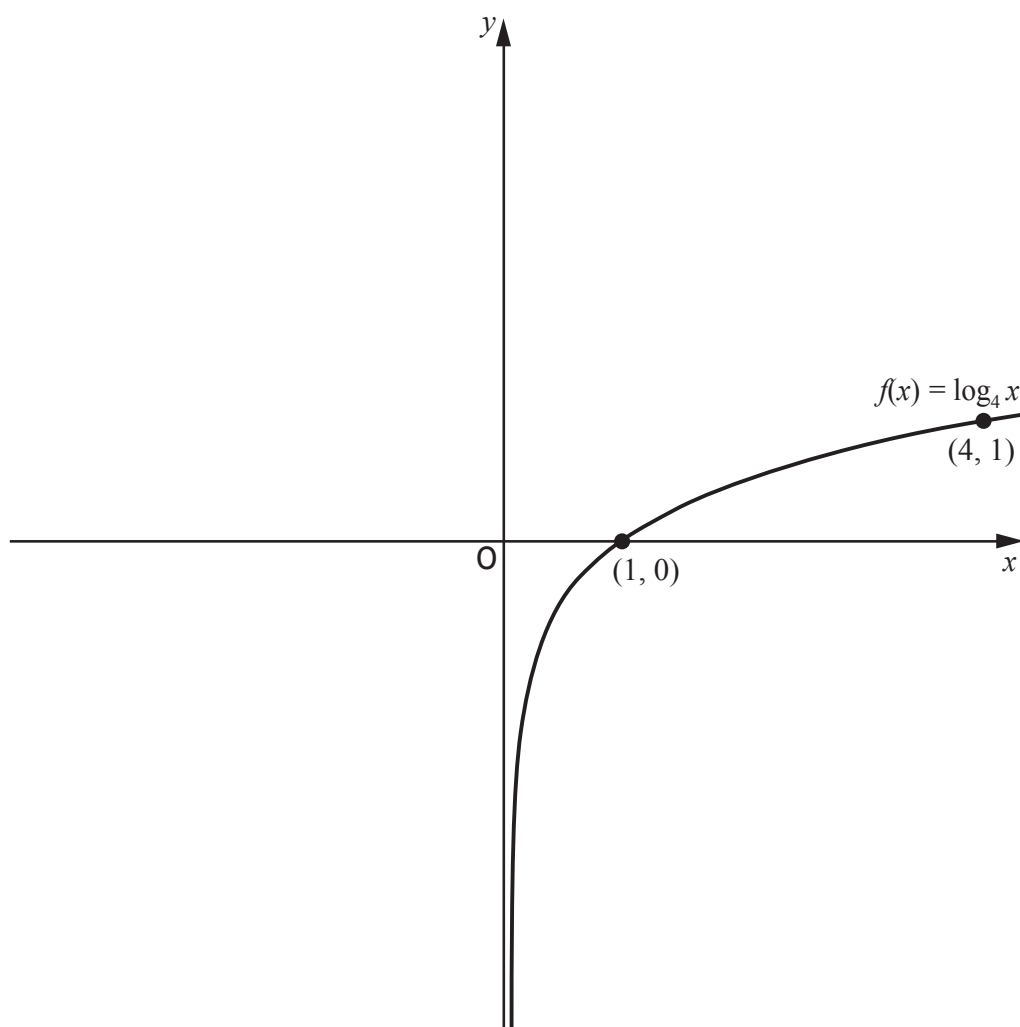
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12.(a)

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National
Qualifications
2016

X747/76/12

**Mathematics
Paper 2**

THURSDAY, 12 MAY

10:30 AM – 12:00 NOON

Total marks — 70

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Scalar Product:

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or $\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$ where $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$.

Trigonometric formulae:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

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Table of standard derivatives:

$f(x)$	$f'(x)$
$\sin ax$	$a \cos ax$
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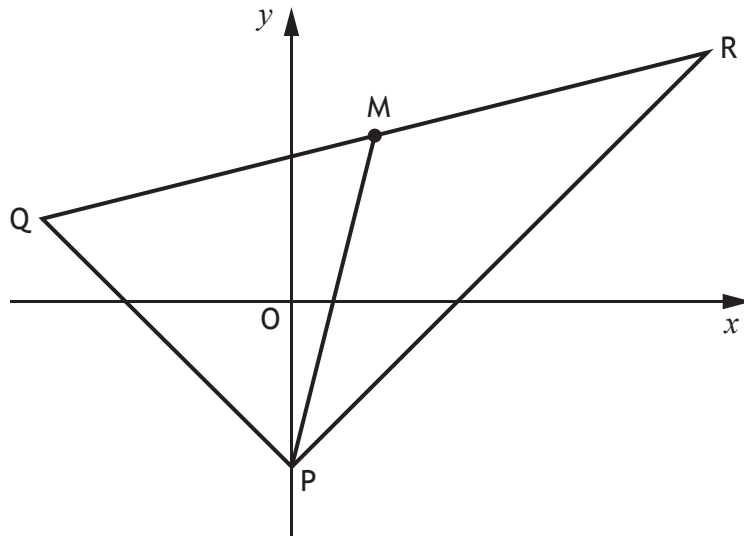
Table of standard integrals:

$f(x)$	$\int f(x) dx$
$\sin ax$	$-\frac{1}{a} \cos ax + c$
$\cos ax$	$\frac{1}{a} \sin ax + c$

Attempt ALL questions

Total marks — 70

1. PQR is a triangle with vertices $P(0, -4)$, $Q(-6, 2)$ and $R(10, 6)$.



- | | |
|---|---|
| (a) (i) State the coordinates of M, the midpoint of QR. | 1 |
| (ii) Hence find the equation of PM, the median through P. | 2 |
| (b) Find the equation of the line, L , passing through M and perpendicular to PR. | 3 |
| (c) Show that line L passes through the midpoint of PR. | 3 |
2. Find the range of values for p such that $x^2 - 2x + 3 - p = 0$ has no real roots.
- 3

[Turn over

3. (a) (i) Show that $(x+1)$ is a factor of $2x^3 - 9x^2 + 3x + 14$.

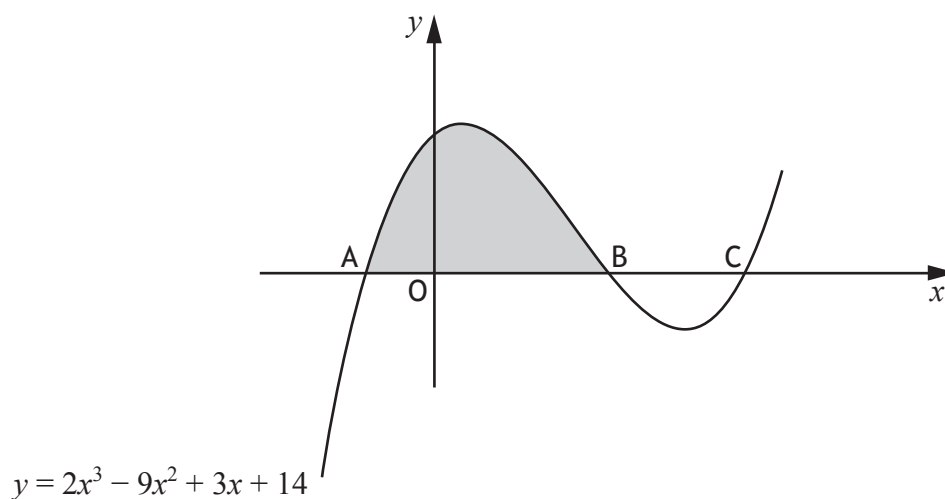
2

(ii) Hence solve the equation $2x^3 - 9x^2 + 3x + 14 = 0$.

3

(b) The diagram below shows the graph with equation $y = 2x^3 - 9x^2 + 3x + 14$.

The curve cuts the x -axis at A, B and C.



(i) Write down the coordinates of the points A and B.

1

(ii) Hence calculate the shaded area in the diagram.

4

4. Circles C_1 and C_2 have equations $(x+5)^2 + (y-6)^2 = 9$ and $x^2 + y^2 - 6x - 16 = 0$ respectively.

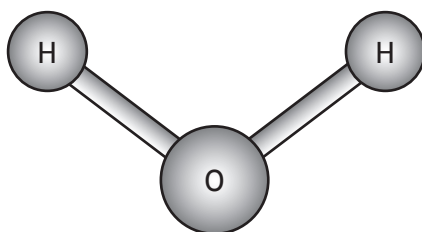
(a) Write down the centres and radii of C_1 and C_2 .

4

(b) Show that C_1 and C_2 do not intersect.

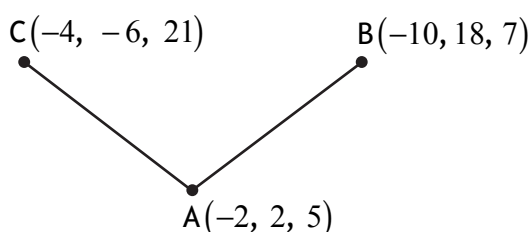
3

5. The picture shows a model of a water molecule.



Relative to suitable coordinate axes, the oxygen atom is positioned at point $A(-2, 2, 5)$.

The two hydrogen atoms are positioned at points $B(-10, 18, 7)$ and $C(-4, -6, 21)$ as shown in the diagram below.



- (a) Express $\vec{AB}$ and $\vec{AC}$ in component form. 2
- (b) Hence, or otherwise, find the size of angle BAC. 4
6. Scientists are studying the growth of a strain of bacteria. The number of bacteria present is given by the formula

$$B(t) = 200e^{0.107t},$$

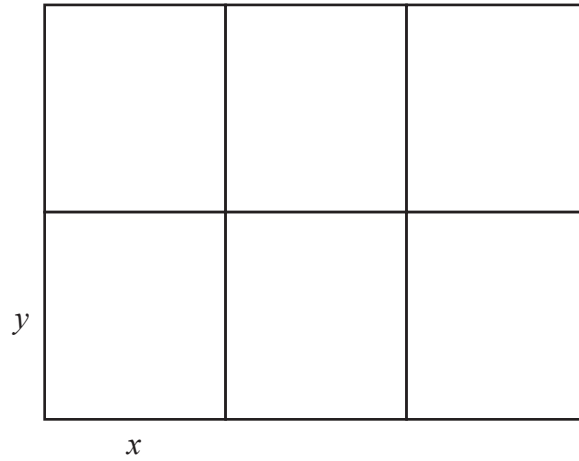
where t represents the number of hours since the study began.

- (a) State the number of bacteria present at the start of the study. 1
- (b) Calculate the time taken for the number of bacteria to double. 4

[Turn over

7. A council is setting aside an area of land to create six fenced plots where local residents can grow their own food.

Each plot will be a rectangle measuring x metres by y metres as shown in the diagram.



- (a) The area of land being set aside is 108 m^2 .
Show that the total length of fencing, L metres, is given by

$$L(x) = 9x + \frac{144}{x} . \quad 3$$

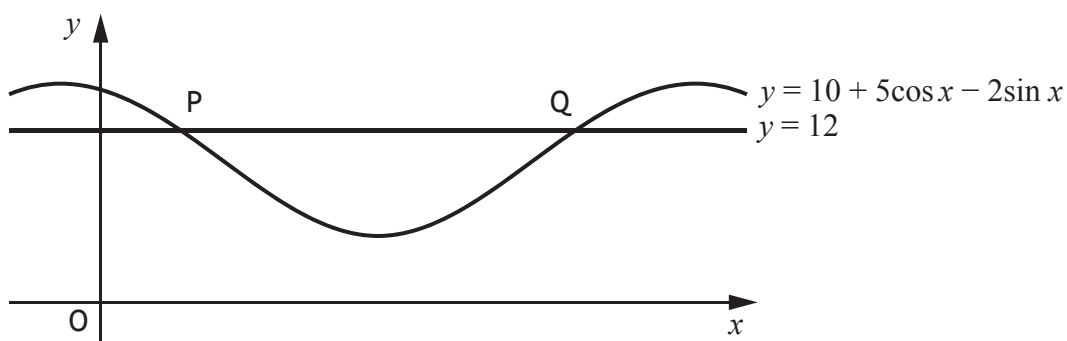
- (b) Find the value of x that minimises the length of fencing required. 6

8. (a) Express $5\cos x - 2\sin x$ in the form $k \cos(x + a)$,
where $k > 0$ and $0 < a < 2\pi$.

4

- (b) The diagram shows a sketch of part of the graph of $y = 10 + 5\cos x - 2\sin x$ and the line with equation $y = 12$.

The line cuts the curve at the points P and Q.



Find the x -coordinates of P and Q.

4

9. For a function f , defined on a suitable domain, it is known that:

- $f'(x) = \frac{2x+1}{\sqrt{x}}$
- $f(9) = 40$

Express $f(x)$ in terms of x .

4

[Turn over for next question]

10. (a) Given that $y = (x^2 + 7)^{\frac{1}{2}}$, find $\frac{dy}{dx}$.

2

(b) Hence find $\int \frac{4x}{\sqrt{x^2 + 7}} dx$.

1

11. (a) Show that $\sin 2x \tan x = 1 - \cos 2x$, where $\frac{\pi}{2} < x < \frac{3\pi}{2}$.

4

(b) Given that $f(x) = \sin 2x \tan x$, find $f'(x)$.

2

[END OF QUESTION PAPER]



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X747/76/02**Mathematics Paper 2
Answer Booklet**

THURSDAY, 12 MAY

10:30 AM – 12:00 NOON



Fill in these boxes and read what is printed below.

Full name of centre

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Town

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Forename(s)

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Number of seat

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Scottish candidate number

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